

Functional

Non Functional

Turn the long URL into Short URL

Redirect to the appropriate link from short URL

User can set the time of expiration for the URL

The Short URL must be unique

Low Latency: The read operation must execute in a few ms. (50ms to 100ms)

Highly Available: The redirection must always be available

URL Shortener: Back of the envelop estimation

To find: Number of Server, Data Storage and RAM for 5 years.

Total Users: 1 Billion

Daily Active User: 200 Million

1 User: 40 Reads, 10 Writes

Total requestes/day: 10 Billion

Requestes/Second: 125

Thousands/second

(Assume) 1 server = 100 thread, Latency = 500ms

Number of Server: 625 Servers

Daily Write: $200M * 10 = 2 \text{ Billion URL/day}$
(Assume) 1URL = 1KB

Daily Storage: 2TB/day

Data Storage(5 years): 3.7PB

Reads per day = 8Billion/day

Total Daily Read Data Volume: 8Billion read* 1 KB/URL = 8 TB/day

Base Daily Hot Cache: 1.6 TB RAM

Project RAM Growth Over 5 Years: 3.3 TB

Cluster Node Count (using standard 64 GB RAM nodes): 106
Cache Servers